

Calibrated X-ray reflectivity transfer across hafnia deposition batches

The NIST source contains 56 measured X-ray reflectivity spectra from twelve hafnia-coated silicon wafers, produced in four deposition batches. The target is native intensity in counts at higher angles; a fixed five-point band near $2\theta=1^\circ$ supplies an explicitly allowed calibration for each spectrum. Whole deposition batches define validation, so positions from a wafer never cross partitions.

The selected finite-film interference surrogate reaches 6,200 counts wafer-balanced MAE on the fourth deposition batch, versus 11,368 for a smooth flexible envelope and 53,216 for the asymptotic Fresnel tail. Its fitted effective period is consistent with an approximately 8.64 nm optical thickness scale. This reproduces established thin-film interference and supports a useful calibrated transfer model; it does not establish a new optical law or certified absolute thickness.

Define $q = 4\pi\sin\theta/\lambda$ with wavelength $\lambda = 1.540593 \text{ \AA}$. The Fresnel intensity factor is $F(q) = [(q - \sqrt{q^2 - q_c^2})(q + \sqrt{q^2 - q_c^2})]^2$. The selected expression below combines this factor with a roughness envelope and film-interference term. Wavevectors have units $\text{\AA}^{-1}$; roughness and thickness are in angstroms, contrast is dimensionless, and phase is in radians. The implementation guards the square root outside its admitted range.

Further fringe decoherence, position-dependent thickness and additive background were tested and preserved. None improved the primary development criterion; several trade small secondary-error differences. One reserved batch is too little for population claims, and correlated parameters prevent a unique structural interpretation.

Experimental scope and evaluation

Predict higher-angle intensity after measuring a fixed low-angle calibration band of the same spectrum. Position and angle are known acquisition settings.

Calibration: Five native intensity samples near 2 theta 1.0 deg per spectrum; no other held-out intensity is a predictor. Same calibration for every model.

Validation unit: Complete deposition batch is the split unit; wafer is the primary aggregation unit. All positions on a wafer stay together. Three deposition batches develop; fourth confirms.

Show each held-out wafer; one reserved deposition batch supplies no independent-batch confidence interval.

Target: native higher-angle XRR intensity (counts). Errors use counts. The selected reference is **kiessig_interference**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
q	$\text{\AA}^{-1}$; $4\pi\sin(\text{theta})/\text{lambda}$
q0	$\text{\AA}^{-1}$; calibration mean wavevector
I0	counts; mean at 2 theta 1.0 ± 0.008 deg
outer	binary non-center measurement position

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
flat_calibration	1.92561e+06	1.94494e+06	2.16329e+06
flexible	10890.6	11368.2	14725.5
fresnel_tail	53063.8	53215.6	61941.9
roughness_envelope	14609.1	14928.4	15641.2
finite_angle_fresnel	10932	11243.9	11337.8
kiessig_interference	5988.58	6199.69	11319.3
contrast_decoherence	6031.67	6257.75	11123.3
center_outer_thickness	6166.24	6266.36	11109.5
instrument_background	6467.69	6693.92	11599.2

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\hat{I}(q) = I_0 \frac{F(q)}{F(q_0)} e^{-\sigma^2(q^2 - q_0^2)} \frac{1 + r \cos(qd + \phi)}{1 + r \cos(q_0 d + \phi)}$$

The variables and physical dimensions are defined above; the guarded executable expression and all alternatives are in EQUATIONS.md.

Coefficient	Frozen value
qc	0.0616271453
sigma	4.56630901
d	86.3623525
r	0.861594601
phase	1.32242034

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- Twelve 200 mm wafers with 65 ALD cycles from one process; three confirmation wafers share one deposition run.
- Intensity-envelope and fringe parameters are effective metrology surrogates; density/thickness/roughness are not independently identified or verified by the author-fitted SE outputs.
- The fixed angle range lies above critical-angle effects; calibrated counts are not absolute reflectivity.

Schema inspection saw first three low-angle counts and last two points near 6 deg in wafer 11, outside the eligible target/calibration bands. Higher-angle eligible confirmation values were not inspected.

Reproduction

Run `python run.py` in this final package to verify hashes and reproduce every saved prediction. Use `python run.py --inputs new.csv --output predictions.csv --model reference` only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: <https://data.nist.gov/od/id/mds2-3930>. See source/units audit and prior-art records in research history.